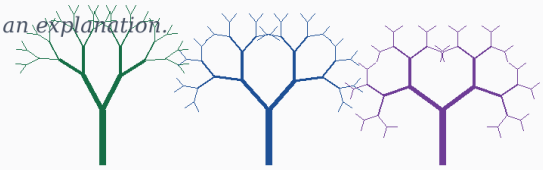
THE CURIOSITY SERIES · ARTICLE IX

The Shape That Keeps Coming Back

Why nature draws the same pattern — and where the pattern stops being an explanation.

Akshat Verma
branching · scaling laws · where analogy ends



A VISUAL INVESTIGATION · THE CURIOSITY SERIES, ARTICLE IX · AKSHAT VERMA

A neuron, a tree, a river delta, the blood vessels in your hand, a galaxy, the cosmic web. Put them side by side and they look like the same drawing. That observation is correct, it is old, and it is genuinely profound. What follows is an attempt to find out how far it goes — which laws are real, which are contested, and at exactly which point a resemblance stops being an explanation and becomes something that cannot be checked at all.

PART I · THE PATTERN IS REAL

Seven pictures, one caption

The claim arrives as a grid of images under a single line: same pattern, different scales, one existence. Before arguing with it, it is worth granting what is true. The shapes really do repeat. Branching is everywhere, at every scale we can observe, and that is not a coincidence or a trick of the eye.

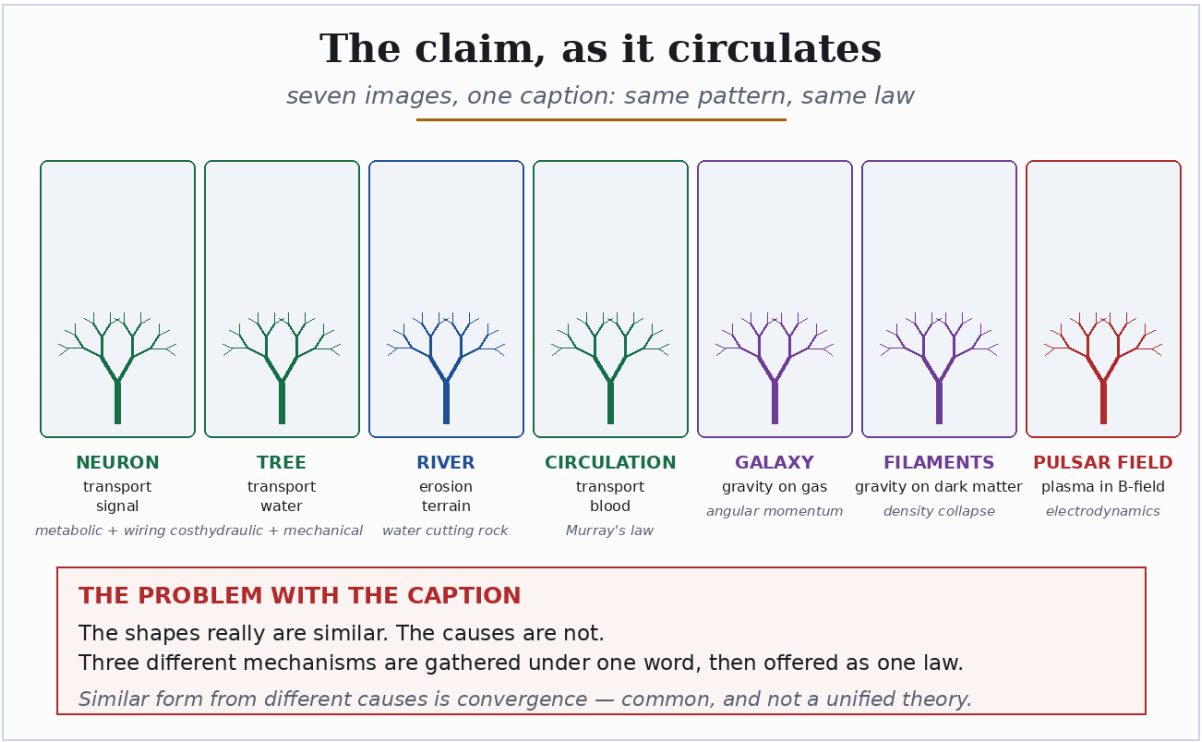


Fig. 1 — The claim as it circulates, with the actual mechanism named under each panel.

But look along the bottom row. The neuron and the vessels move a substance. The river cuts rock. The filaments are gravity acting on dark matter over billions of years. Three different physical processes have been collected under one word, and the resemblance is then offered as evidence of a single law. Whether that inference holds is the question this article exists to answer.

There is a real law here — and it is narrower than advertised

In 1926 Cecil Murray asked what the best possible branching geometry for a blood vessel would be. Wide tubes are cheap to push fluid through but expensive to fill with blood and keep alive. Narrow tubes are the reverse. Minimise the sum of those two costs and an exact rule falls out.

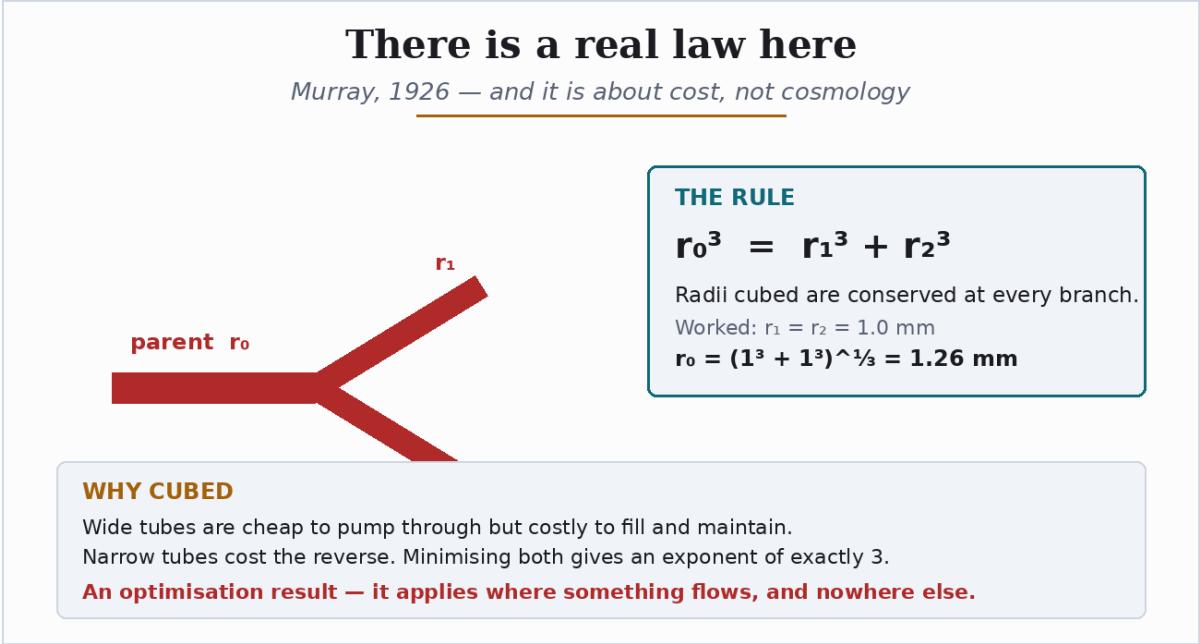


Fig. 2 — Murray's law. Radii cubed are conserved at every bifurcation.

Murray's law is real, measurable, and confirmed in most of the vasculature below the largest arteries. It also holds in plant xylem and insect tracheae. It is a beautiful result. And notice precisely what it requires: something must flow, and two competing costs must be paid. Where those conditions are absent, the law says nothing at all.

So why do rivers and galaxies look the same as lungs?

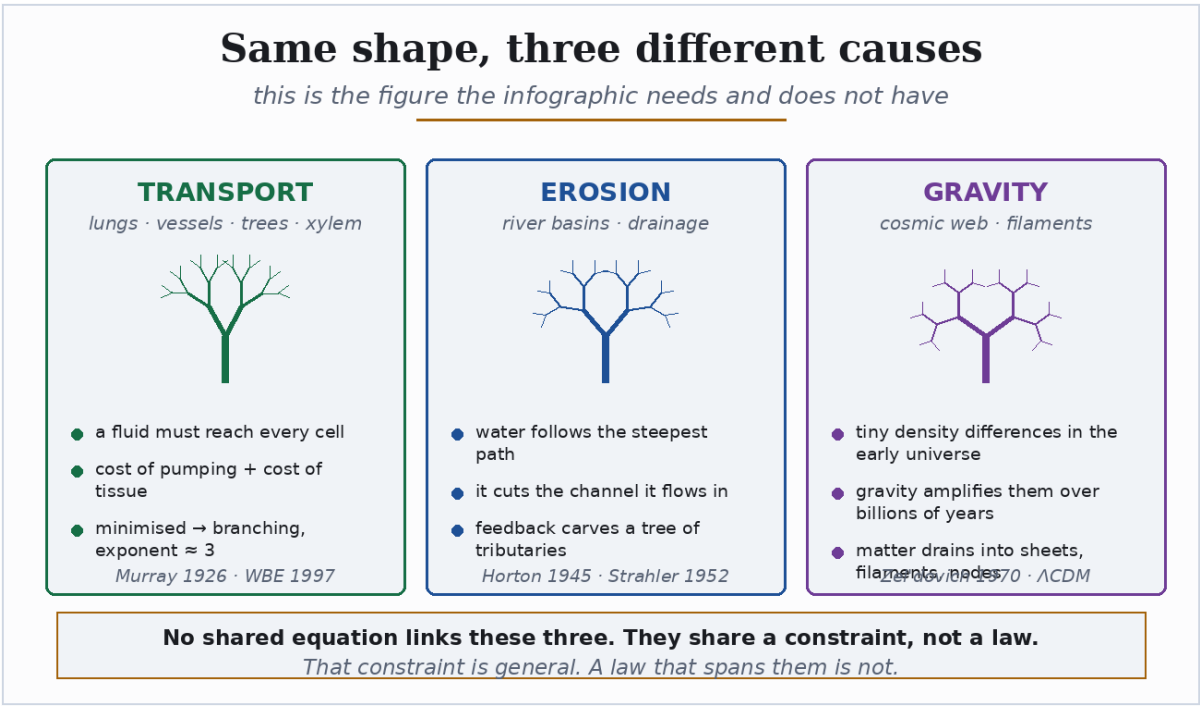


Fig. 3 — Same shape, three different causes. This is the figure the poster needs and does not have.

Rivers branch because water takes the steepest path and erodes the channel it travels in — a feedback between flow and terrain, with no cost function being minimised by anything. The cosmic web branches because tiny density variations in the early universe were amplified by gravity until matter drained into sheets, filaments and nodes. Nobody is optimising. There is no transport problem being solved.

The word that should replace "unified theory". **Convergence**. Different mechanisms producing similar form because they face a similar geometric constraint — here, fill a volume from a limited number of sources. Constraints that general will produce similar-looking answers in unrelated systems. That is interesting, and it is the opposite of evidence for one underlying law.

PART II · WHERE THE PATTERN BECAME A NUMBER

What happens when you try to measure it?

The strongest version of this idea is not a picture. It is a number. Metabolic rate does not scale in proportion to body mass; it scales as mass to the power of about three quarters, and it does so across animals spanning many orders of magnitude. Kleiber found it in 1932.

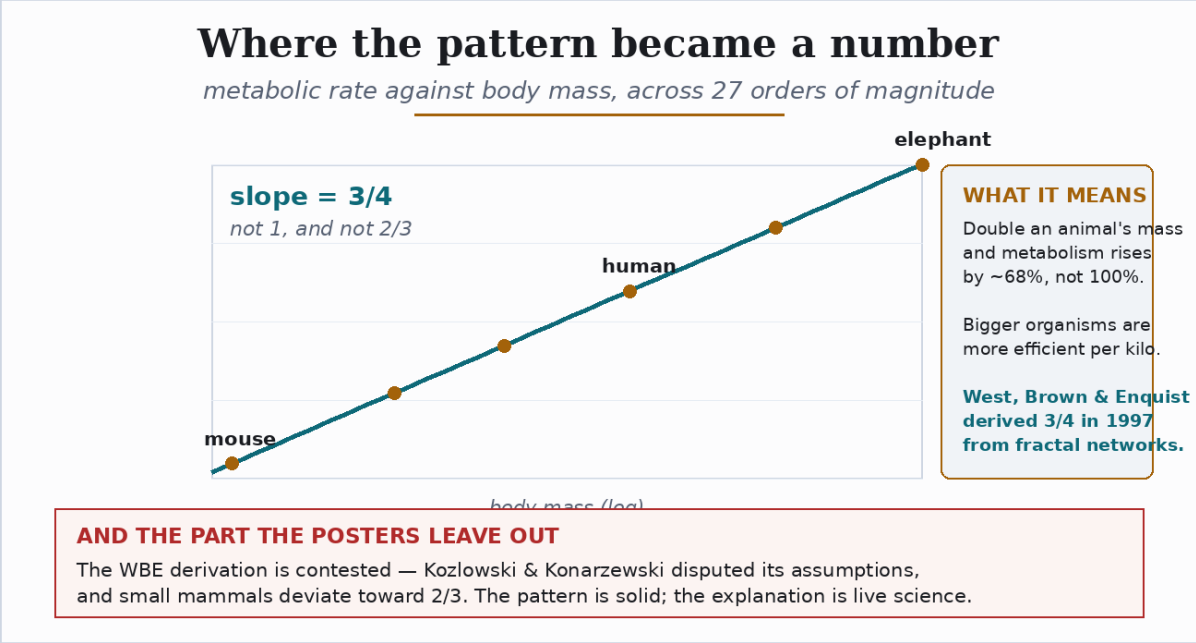


Fig. 4 — Kleiber's law, and the part the posters leave out.

In 1997 West, Brown and Enquist derived that exponent from exactly the idea this article started with: organisms are supplied by space-filling fractal branching networks, and if those networks minimise dissipated energy while terminal units stay a fixed size, three quarters falls out of the mathematics. This is the real version of "one principle across scales," and it was published in Science.

And here is the discipline the infographic skips. The WBE derivation is contested. Kozlowski and Konarzewski challenged its assumptions; Etienne and colleagues published a paper whose title begins "Demystifying"; small mammals deviate toward two thirds; later work argues the derivation is structurally divergent at the very exponent it assumes. The *pattern* is solid. The *explanation* is live, unfinished science. Citing it as settled is the same error as citing the picture as proof.

PART III · WHERE THE PATTERN STOPS EXPLAINING

How far does the idea actually reach?

Homeostasis is next in the chain: living systems resist instability, regulate themselves, return toward balance. This is real physiology and it is well characterised. But it is a control mechanism operating inside an organism, built

from feedback loops with sensors and effectors. It describes how a body maintains a state. It is not a property of matter, and it does not scale upward to physics.

The poster then takes one more step, and it is the step where the argument leaves science entirely: if life resists ending, perhaps existence itself does, and perhaps consciousness continues along a branch where survival remained possible.

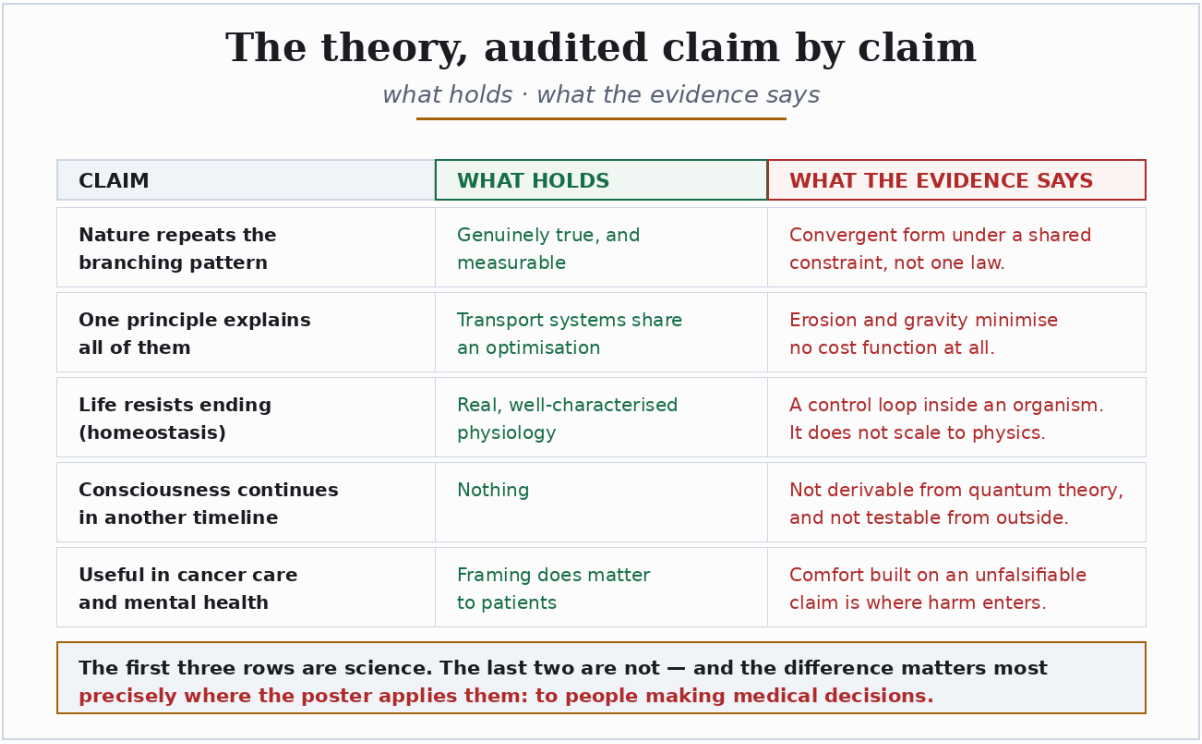


Fig. 5 — The theory audited claim by claim.

What is wrong with that last step?

Not that it is strange. Physics is full of strange things that turned out to be true. The problem is structural: no observation could ever contradict it. In any branch where you did not survive, there is no you to notice. The claim is therefore compatible with every possible experience, which means no experience can count as evidence for or against it.

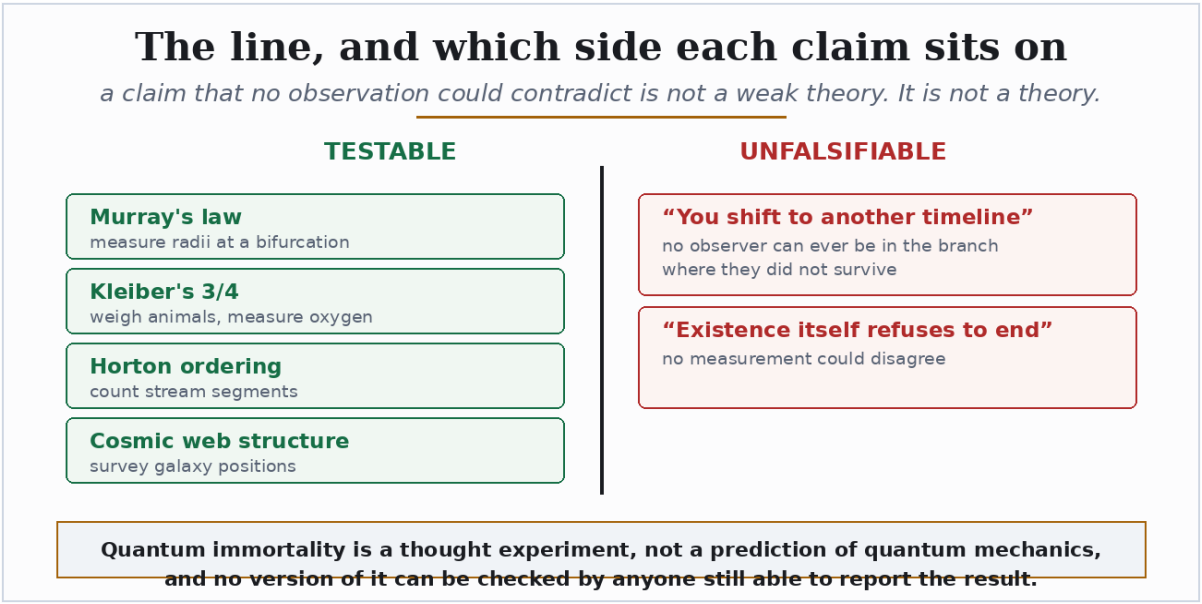


Fig. 6 — The line, and which side each claim sits on.

Quantum immortality is a known thought experiment, usually discussed as a probe of what the many-worlds interpretation would imply, not as a prediction of quantum mechanics. Quantum mechanics does not say that consciousness selects a branch. The mathematics contains no term for an observer's continuity. The idea is a philosophical extrapolation wearing the vocabulary of physics.

Why be strict about this one?

Because of where the poster applies it. The use cases listed include cancer care and mental health: reduce fear of death, help patients focus on quality of life, support resilience. Framing and meaning genuinely do matter to people who are ill — that part is not in dispute.

But this is the line that matters. Comfort built on an unfalsifiable claim is fragile in exactly the moment it is needed, and it can shift real decisions. "You do not experience the outcome where you die" is not a neutral reassurance offered to someone weighing treatment. Anything presented to a patient as a reason to fear death less should be able to say what would show it to be wrong. This one cannot.

PART IV · THE VERDICT

What survives

A great deal, and it is more interesting than the version it replaces. Branching really does recur across biology, geology and cosmology. Within transport systems there really is a shared optimisation, with an exact exponent that has been measured for a century. Kleiber's law really does hold across twenty-seven orders of magnitude, and the attempt to derive it from branching networks is serious science that is still being argued over today.

What does not survive is the leap from that to a single law of existence. The pattern recurs because the constraint recurs, not because one principle is running underneath everything. And the final step — from homeostasis to consciousness persisting across timelines — is not a bolder version of the same argument. It is a different kind of claim, one that no measurement could ever touch.

What I would tell myself. A resemblance is a question, not an answer. When two systems look alike, the useful move is to ask what each is actually doing — and if the mechanisms differ, the similarity is a fact about constraints, not about a hidden unity. And when a claim starts offering comfort to people making medical decisions, the standard for it should go up, not down.

SOURCES · Murray, PNAS 12:207 (1926) · Kleiber (1932) · West, Brown & Enquist, Science 276:122 (1997) · Kozlowski & Konarzewski (2004) · Etienne, Apol & Olff, Functional Ecology 20:394 (2006) · Horton (1945), Strahler (1952) · Zel'dovich (1970) and ΛCDM structure formation. Figures original; each states what it simplifies.